

From Hostility to Deliberation: A Sociotechnical Model for Mitigating Unproductive Debates in AI-Mediated Social Media

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Abstract. Civil society’s participation in digital public debate is undermined by the operational logic of mainstream platforms, whose algorithms prioritize engagement over the quality of discussions, fostering affective polarization and unproductive debates. Given the absence of design mechanisms capable of mitigating these dynamics, we propose a sociotechnical model, informed by Knowledge Management principles, to support the identification, organization, and mitigation of unproductive debates. This qualitative, documentary research is based on a comparative analysis of the Reddit, Discord, Mastodon, Bluesky, and Kialo platforms. From this synthesis, articulated with Marc Angenot’s theoretical framework, we derive a model comprising five mitigation dimensions cut across by a Generative AI mediation layer, under the principle of human–AI symbiosis. An illustrative application to a real-world case demonstrates the model’s potential to restore the conditions for deliberation and collective knowledge creation, offering guidance to platform designers and managers.

Keywords: Unproductive Debates · Affective Polarization · Sociotechnical Design · Generative AI · Social Media Platforms

1 Introduction

Civil society’s participation in contemporary democracy is affected with respect to the discussions and content produced within social media [25]. Although these tools have the potential to foster democratic engagement, they frequently become vectors of extreme polarization and unproductive debates [11, 20], resulting in greater hostility between antagonistic groups and highlighting the critical need to study how digital architectures can be redesigned to promote healthier and higher-quality public debates.

Polarization is inherent to social dynamics and, in many contexts, proves healthy and even necessary for the plurality of ideas, critical thinking, and the exercise of democracy [16, 25]. The central problem lies in the operational logic of mainstream platforms, whose algorithms and features, such as recommendation and moderation, are predominantly oriented toward metric-driven engagement and user retention, to the detriment of the quality of debates and of the knowledge generated from them [1, 25] — a configuration we refer to throughout this

work as the *engagement logic*. Added to this is the absence of sociotechnical mechanisms — of interface, interaction, and governance — capable of identifying and surfacing the dynamics that generate unproductive debates, which severely limits the productivity of discussions [25].

To address this gap, this work proposes a model oriented toward the design of digital platforms that supports the identification, organization, and mitigation of unproductive debate dynamics. The model is built from a comparative analysis of mechanisms present in alternative platforms and is informed by Knowledge Management (KM) principles, articulating affordances for argumentative structuring, curation, and synthesis of collective knowledge.

Additionally, this research explores the emerging role of Generative Artificial Intelligence (AI) and Large Language Models (LLMs) as support for KM processes [8], investigating how these technologies can act as mediation tools to analyze discourse, organize and interpret collective knowledge, and transform unproductive interactions into more robust deliberative processes.

Research Questions To guide the development of this study, the following research questions were defined:

- RQ1: What mechanisms do alternative platforms offer for mitigating unproductive debates?
- RQ2: How can a sociotechnical design model, supported by Generative AI and informed by KM principles, support the identification, organization, and mitigation of unproductive debates?

This work provides guidance to designers of sociotechnical systems, governance managers, and researchers in digital democracy and KM, through a model that indicates characteristics and mechanisms capable of fostering productive debates.

The remainder of this paper presents the related work and the theoretical background (Sections 2 and 3), the methodology and corpus (Section 4), the categories of mechanisms and the comparative analysis (Sections 5 and 6), the model and its application to an empirical case (Sections 7 and 8), and, finally, the discussion and conclusions (Sections 9 and 10).

2 Related Work

The mitigation of unproductive debates and polarization on digital platforms has been the object of growing interest in Human-Computer Interaction (HCI) and Computer-Supported Cooperative Work (CSCW). This section reviews the literature along three fronts aligned with the scope of this research: decentralized architectures, interface design, and the use of AI to support moderation and digital debates.

In contrast to the engagement logic of mainstream networks, new decentralized and structured architectures have been proposed [17, 28]. The work of

Schneider et al. [25] is particularly central to this research, as it distinguishes affective polarization, a systemic and destructive risk, from legitimate ideological conflict. Based on a comparative analysis of platforms such as Mastodon, Bluesky, Kialo, Reddit, and Discord, the authors demonstrate that the mitigation of extreme polarization emerges from specific sociotechnical affordances, such as decentralization, algorithmic composability, and interactional structuring, which reduce systemic toxicity and limit affective escalation while preserving space for productive disagreement.

Complementarily, Jeong et al. [9] offer a comprehensive taxonomy of Decentralized Online Social Networks (DOSNs), categorized into federated, peer-to-peer, blockchain-based, and hybrid models, and correlate these architectural choices with persistent social challenges on the web, such as social fragmentation and inconsistencies in content moderation.

2.1 Interface Design and Feed Curation

Regarding interface design and feed curation, Nelimarkka et al. [17] show that (re)design inspired by Habermasian ideals of the public sphere, with recommendations that diversify sources and promote role-taking, can lead to depolarization, albeit at the cost of a more complex user experience. Liu et al. [15], in turn, examine decentralized feed curation on Mastodon and reveal that seamless design approaches, with visible seams in the interface, increase trust in algorithmic curation, highlighting the trade-offs between rule-based and machine-learning-based filtering.

2.2 Artificial Intelligence and Argumentative Assessment in Moderation

As for the role of Generative AI as mediation infrastructure, Zhang et al. [28] map the attitudes of operators of decentralized social networks, who categorically reject AI as an autonomous decision-making actor in moderation and envision it instead as collaborative governance infrastructure, under strict boundaries that prioritize human accountability, reversibility of actions, and data transparency.

In summary, the reviewed literature shows advances in the understanding of decentralized architectures, in the (re)design of interfaces for depolarization, and in the use of LLMs to support moderation. However, these efforts remain fragmented: they analyze isolated mechanisms or specific platforms without articulating them into an integrated model to guide the design of platforms aimed at mitigating unproductive debates. This work seeks to fill this gap by proposing a model that integrates affordances for deceleration, structuring, curation, exposure, and moderation, cut across by a Generative AI mediation layer.

3 Theoretical Background

This section presents the theoretical foundations underpinning this research, articulating three complementary axes. First, unproductive debate is characterized

as a social phenomenon based on Marc Angenot’s analytical categories, followed by a discussion of how the architectures of mainstream social media act as catalysts of these dynamics. Next, the phenomenon is examined through the lens of collective knowledge creation, showing why unproductive debates undermine social learning processes on digital platforms. Finally, we discuss the role of Generative AI as a sociotechnical mediator capable of supporting the organization and qualification of interactions.

3.1 Unproductive Debate as a Social Phenomenon

To understand the dynamics of interactions in the digital environment, it is first necessary to recognize that argumentative failure is not a phenomenon exclusive to the internet [4]. In his analyses of the history of ideas, Marc Angenot argues that mutual incomprehension in debates does not simply derive from the participants’ ignorance, lack of logic, or bad faith, but rather from a structural condition of social life [3, 4]. Angenot characterizes this phenomenon as a “dialogue of the deaf,” a rule of modern social life that renders many debates structurally unresolvable [4]. From this perspective, unproductive debate can be understood and unfolded through three central analytical categories, which will appear in italics: *discursive incommensurability*, *rhetoric of incomprehension*, and *illusion of rationality*. The first of these is *discursive incommensurability*, which occurs when participants operate in incompatible “discursive universes” or “ideological languages,” not sharing the same values, presuppositions, or criteria about what constitutes truth [3].

The second category is the *rhetoric of incomprehension*, which concerns the refusal or inability to recognize the adversary as a rational being within their own system of thought [4]. Instead of promoting a productive confrontation, digital debate quickly degenerates into attack, irony, caricature, and delegitimization of the opponent — a scenario in which the divergent position is simply labeled as irrational nonsense.

Coupled with this is the *illusion of rationality*, which consists of the unrealistic expectation that argumentative debate naturally tends toward rational consensus or that it is actually possible to “convince the other” [4]. In practice, individuals argue more to justify themselves before their own audience than to change others’ opinions, resulting in parallel monologues and repetitive cycles of discussion. This is, therefore, a dynamic regulated by the social discourse of an era itself, which establishes the hegemonic limits of what is sayable and opinable [5].

3.2 How Mainstream Social Media Amplify the Problem

If unproductive debates were already an inherent feature of analog debates, the architectures of digital social networks have acted as strong catalysts of this problem: by introducing new speeds and metrics for communication, they substantially altered the way communities form and share information [10, 25], and the engagement logic intensified segregation into echo chambers [20].

Consequently, the virtual environment came to harbor and amplify resentful and denialist discourses, in which emotions such as indignation and hatred are mobilized to annihilate any opposing view [14]. In these spaces, discursive construction rests on victimization and the rejection of science, creating bubbles that reject dialectics and hinder the mediation of democratic conflicts through productive debates.

3.3 Collective Knowledge Creation and Generative AI

To reverse these dysfunctions, it is necessary to see social media not merely as communication channels, but as sociotechnical systems in which collective knowledge creation takes place, whose functional building blocks regulate the speed, direction, and nature of informational exchange [10, 19]. Through the lens of Nonaka’s dynamic theory of knowledge creation [18], knowledge is generated by a continuous spiral of conversion between tacit and explicit knowledge, expressed in the SECI model (Socialization, Externalization, Combination, and Internalization). Unproductive debates block this spiral above all in the first two phases: socialization, which requires mutual trust and the sharing of authentic experiences, is rendered unfeasible by the *rhetoric of incomprehension* and by *discursive incommensurability*; and externalization, which depends on continuous dialogue to articulate tacit knowledge into shareable concepts, collapses under the *illusion of rationality* and the delegitimization of the opponent. Once the spiral is broken, debate degenerates into parallel monologues, fragmenting the network and generating stagnation.

In this scenario, KM gains a powerful ally in Generative AI, provided it operates under the principle of human–AI symbiosis rather than full automation or replacement of the human actor [8]: AI takes on exhausting tasks of capturing, organizing, and filtering large volumes of information, reducing cognitive overload and freeing participants for analytical, critical, and deliberative activities [27]. In digital debates, this means synthesizing content, extracting topics, and identifying divergences in a structured way, organizing the interactional chaos without replacing human decision-making; the specific roles of this mediation in each dimension of the model are detailed in Section 7.6.

4 Methodology

This research is characterized as qualitative, comparative, documentary, and theoretical-conceptual, with the objective of analyzing how alternative digital platforms can support the mitigation of unproductive debates through design and interaction mechanisms. The methodological path comprised four steps: (i) a literature review on digital debate, its moderation and synthesis mechanisms, and KM approaches supported by Generative AI; (ii) documentary analysis of the selected platforms; (iii) elaboration of a comparative matrix of their mechanisms, guided by the categories defined in Section 5; and (iv) synthesis of the

most promising mechanisms, from which the proposed model and its illustrative application to an empirical case are derived.

The selection of platforms is based on purposive sampling by theoretical relevance, favoring cases with significant architectural and functional diversity for the problem under study, according to five criteria: (i) degree of centralization or decentralization of the network; (ii) presence of public or semi-public debate; (iii) existence of distinct moderation and governance mechanisms; (iv) provision of specific argumentative structures; and (v) availability of public technical documentation.

Based on these criteria, Mastodon, Bluesky, Kialo, Reddit, and Discord were selected, as they represent contrasting paradigms of sociotechnical mediation [25]. Mastodon and Bluesky embody the decentralized social network model, in which federation and instance autonomy mitigate engagement-logic algorithms [9]; Kialo offers an architecture devoted exclusively to structured debate in argument trees, a counterpoint to the linearity of traditional timelines; and Reddit and Discord operate governance logics based on subcommunities and volunteer moderation [25]. Mainstream platforms such as X and Facebook were deliberately excluded because their very architecture constitutes the vectors of the problem the model aims to mitigate [20, 25].

5 Mechanisms for Mitigating Unproductive Debates

Given the diagnosis of “dialogues of the deaf” [4], the main way forward drawn from Angenot is to shift the goal of the confrontation: instead of immediate persuasion, which runs up against the *illusion of rationality*, the aim is to understand the other’s system of thought. We operationalize this proposition as meta-communication, a term Angenot does not formally employ: communicating about the communication itself, making explicit the hidden presuppositions, implicit rules, underlying values, and what each side considers valid evidence, so that participants can see the internal coherence of the adversary’s reasoning even when they disagree with it. Based on this framework, we define five categories of mitigation mechanisms, which guide the comparative analysis:

- **Debate Deceleration Mechanisms:** interface frictions that reduce impulsiveness and the pressure for immediate positioning, restoring the cognitive time needed for reflection.
- **Divergence Display Mechanisms:** visual tools that synthesize points of consensus and dissent, shifting the goal from persuasion to mapping the differences in rationality among debaters.
- **Debate Structuring Mechanisms:** forms of organizing interactions that make explicit the rules and presuppositions within which the debate is situated even before it begins.
- **Context and Traceability Mechanisms:** resources for preserving the collective memory of the debate that shift the focus from raw content to the “conditions of the dialogue.”

- **Systemic and Algorithmic Moderation Mechanisms:** moderation layers, human or automated, capable of detecting evidence of unproductive debates and redirecting the interaction toward more productive forms, rather than merely removing content.

The subsequent analysis investigates whether and how the selected platforms implement these categories in practice and to what extent they use, or could use, Generative AI to support them.

Table 1. Comparative analysis of platforms and their mechanisms for mitigating unproductive debates.

Platform	Debate Deceleration	Divergence Display	Context and Traceability	Debate Structuring	Systemic and Algorithmic Moderation
Discord	Partially implements, predominantly human intervention, quantitative	Analogous functionality, predominantly human intervention, implicit	Partially implements, predominantly human intervention, partial	Directly implements, hybrid use, high	Directly implements, hybrid use
Reddit	Analogous functionality, predominantly human intervention, quantitative	Does not implement, not applicable, not implemented	Partially implements, predominantly human intervention, partial	Directly implements, hybrid use, high	Directly implements, hybrid use
Bluesky	Does not implement, not applicable, not applicable	Does not implement, not applicable, not implemented	Partially implements, hybrid use, partial	Partially implements, hybrid use, medium	Directly implements, hybrid use
Mastodon	Partially implements, predominantly human intervention, quantitative	Does not implement, not applicable, not implemented	Analogous functionality, predominantly human intervention, ephemeral	Partially implements, predominantly human intervention, low	Partially implements, predominantly human intervention
Kialo	Analogous functionality, predominantly human intervention, qualitative	Directly implements, predominantly human intervention, explicit	Directly implements, predominantly human intervention, persistent	Directly implements, predominantly human intervention, high	Partially implements, predominantly human intervention

6 Analysis of the Alternative Platforms

The classification evaluates each mechanism along three dimensions, synthesized in Table 1: *functionality* indicates the degree of adherence to the theoretical concept (Directly implements, Partially implements, Analogous functionality, or Does not implement); *approach* distinguishes the role of the agents (predominantly AI/Bot use, predominantly human intervention, hybrid use, or not applicable); and the *specific classifications* detail the operation of each mechanism: Quantitative deceleration (limits and temporal barriers) or Qualitative deceleration (encouragement of reflection); Explicit divergence display (formal visualizations) or Implicit (inferred from the discursive structure); High, Medium, or Low structuring, according to the robustness of the thematic structures; and Persistent, Partial, or Ephemeral context, according to the retention of collective memory. In the following subsections, the analysis focuses on what the table does not express: the qualitative limitations of each platform and the role AI plays in it.

6.1 Discord

Discord’s architecture of channels, threads, and forums reduces informational chaos, but depends heavily on manual configuration by moderators. Its frictions, such as slow mode, are oriented toward systemic safety and act as volume reducers, not as mediators of depth, and the platform lacks native knowledge repositories, such as integrated wikis. AI, present in AutoMod, detects spam and toxic content in real time as a binary-logic “janitor”: it removes obvious violations but does not interpret ambiguities.

6.2 Reddit

Reddit has the most robust infrastructure for making presuppositions and rules explicit before the debate, via subreddits, flairs, and wikis. However, its voting system can hide legitimate minorities in favor of popularity, fostering parallel monologues, and its moderation tends to be reactive and restrictively phrased [12], privileging content removal over redirecting the interaction. In moderation, the AI AutoModerator uses LLMs to analyze posts according to the rules of each subreddit, increasing the consistency of decisions, together with spam filters and Crowd Control.

6.3 Bluesky

Bluesky transfers meta-communication to the community: community labeling, algorithmic choice of feeds, and composable moderation form a distributed governance. This logic, however, depends on developers to make technical data visible to the ordinary user, inherits the fast-paced aesthetics of mainstream networks, and may, paradoxically, create bubbles in which disagreeing groups silence the debate. AI is architecturally encouraged: independent agents employ it in feed curation and in the autonomous generation of moderation labels.

6.4 Mastodon

On Mastodon, friction occurs in the consumption of information, not in its production: impulsive replies remain possible, chronological interactions are ephemeral, and moderation depends excessively on humans. Although exemplary in community self-determination, its moderation is binary — dialogue is accepted or dissent is removed via instance blocking — which can generate bubbles and a perception of censorship [6]. There is no evidence of AI use in the analyzed documentation; all automation is based on simple matching rules, leaving governance as a strictly human task.

6.5 Kialo Edu

Kialo Edu operationalizes the disruption of the *illusion of rationality*: its argument trees force divergent premises to inhabit the same visual space, and its

moderation acts on the conditions of the dialogue before the argumentation. Its limitations stem from this rigor: continuous human approval creates bottlenecks and fatigue among administrators, and the interface demands high information literacy, potentially restricting the user profile. AI is not used; the system is grounded in interface design and participatory human curation, with evident potential for future AI-augmented implementations.

6.6 Synthesis of Findings

In summary, debate structuring and systemic moderation are the most consolidated mechanisms, especially on centralized platforms, while qualitative deceleration and divergence display remain restricted to Kialo Edu, and AI, where present, acts as a reactive “janitor” rather than as a mediator of the debate. These gaps, revisited in the Discussion, underpin the five dimensions of the model proposed in the next section.

7 A Sociotechnical Model for Mitigating Unproductive Debates

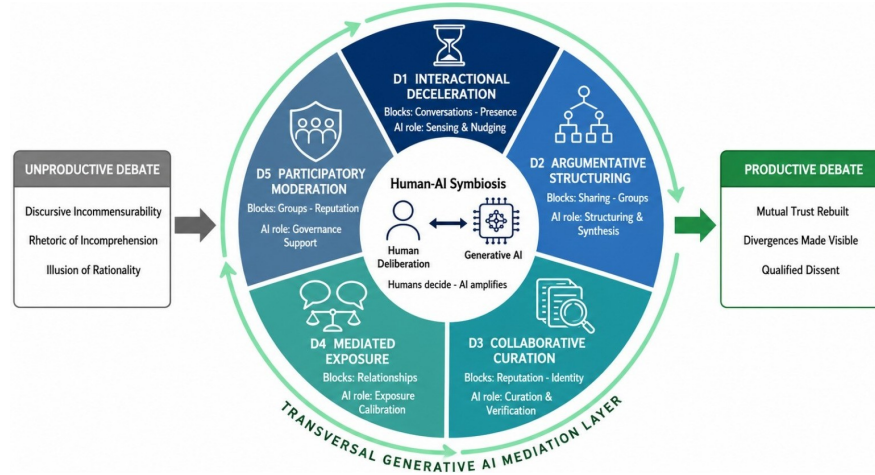


Fig. 1. The five mitigation dimensions, the Honeycomb functional building blocks each one mobilizes, and the transversal Generative AI mediation layer.

The model aims to provide theoretical and practical foundations for the design and evaluation of digital platforms that foster productive debates, starting from the assumption that social media operate as sociotechnical ecosystems of collective knowledge creation [7,23], in which technological affordances structure how individuals interact and organize themselves [13].

The choice of the Honeycomb [10] as the architectural basis is justified by its operationalizability: while the affordances perspective describes emergent relations that are difficult to translate into design prescriptions, the Honeycomb offers a discrete functional decomposition of the social experience, suitable for deriving design dimensions. It should be noted that it was conceived as a descriptive tool, aimed at understanding how organizations engage audiences; here, each functional block is repurposed as a design lens for debate productivity.

The Honeycomb defines social media through seven functional building blocks — Identity, Conversations, Sharing, Presence, Relationships, Reputation, and Groups — whose balance dictates the informational dynamics of the platform; each block is characterized further on, in the dimension that mobilizes it. These blocks are not mutually exclusive [10]: the same block can be mobilized by more than one dimension under distinct facets, such as Reputation, invoked both in content credibility and in conduct sanctioning. By convention, the blocks are written in italics with an initial capital letter, and the dimensions are referred to by the labels D1–D5. Figure 1 shows the orchestration of the blocks, the dimensions, and the transversal Generative AI layer.

When configured under the *engagement logic*, these blocks generate a hostile environment in which *discursive incommensurability* and the *rhetoric of incomprehension* prevail. To reverse this dynamic, the model reconfigures the social functions through five mitigation dimensions, which unite the empirical findings of the comparative analysis with the support of Generative AI: Interactional Deceleration (D1) acts on the pace of interaction; Argumentative Structuring (D2), on the organization and visualization of the debate; Collaborative Curation (D3), on the provenance and persistence of evidence; Mediated Exposure (D4), on contact with divergent perspectives; and Participatory Moderation (D5), on the governance of conduct. Read in light of the SECI spiral, D1 reestablishes the trust conditions of Socialization, D2 supports Externalization, D3 operates Combination, D4 fosters Internalization, and D5, rather than corresponding to an isolated phase, sustains the governance conditions that keep the cycle operating.

7.1 Interactional Deceleration (D1)

This dimension acts on the pace of interaction, not on the organization of content (D2). It mobilizes *Conversations*, which on mainstream networks operate at very high speed, and *Presence*, whose constant display induces an expectation of availability and impulsive replies [10, 19].

The model proposes mitigating this urgency through architectural frictions, such as those observed on Kialo, Discord, and Reddit, which, by restoring cognitive time and reducing reactive hostility, restore the conditions of mutual trust necessary for dialogue. The AI layer acts here as sensing, monitoring the speed and “temperature” of *Conversations* and triggering deceleration nudges upon detecting affective escalation [6].

7.2 Argumentative Structuring (D2)

This dimension concerns the organization and visualization of argumentative content in an ongoing debate, not the entry of external perspectives into the user’s network (D4). It mobilizes *Sharing*, whose unrestricted flow in linear timelines favors informational chaos and parallel monologues [10], and *Groups*, communities that define the valid formats and rules of contribution.

The model postulates visual and structural mechanisms that make the debate’s presuppositions explicit, such as Kialo’s pro/con trees, forcing the typification of posts and countering the *illusion of rationality*, in addition to divergence display mechanisms that enable the mapping of the other’s belief system. The AI layer takes on structuring and synthesis, categorizing premises and visually summarizing each point being defended, so that users can see the coherence of the adversary’s logic without the noise of direct attacks.

7.3 Collaborative Curation (D3)

This dimension deals with the provenance and persistence of evidence, qualifying content rather than sanctioning conduct (D5). It mobilizes *Reputation*, in its content-credibility facet, and *Identity*, in the authorship of cited sources.

The model postulates a verifiable and traceable collective memory repository, shifting the focus from raw content to the conditions of the dialogue. It is precisely in the synthesis and indexing of knowledge from dispersed sources that the literature locates the greatest potential of Generative AI as a knowledge co-creator [2]: the AI layer performs curation and verification, indexing evidence and historical contexts and qualifying the credibility of content before its dissemination [19, 24].

7.4 Mediated Exposure to Divergent Perspectives (D4)

This dimension regulates the entry of external perspectives into the user’s feed, not the organization of an already established debate (D2). It mobilizes *Relationships*, an intermediate level of interaction between the individual and the collective, necessary for the joint creation of knowledge [26]. There is, however, an inherent tension: while homophily fosters echo chambers, abrupt exposure to strongly divergent perspectives produces cognitive resistance and reinforces preexisting beliefs — mere diversity of opinions does not guarantee openness to dialogue [1, 21].

The model postulates that the depolarizing effect depends on calibrating exposure — ideological distance, frequency, and gradualness — exemplified by the insertion of content in “homeopathic doses” [22]: a pendular movement between informational comfort and mild dissent which, in line with Angenot, does not aim at persuasion but at controlled exposure to the internal coherence of the other’s thinking. The AI layer mediates exposure, mapping the polarity of *Relationships* and progressively introducing cognitively moderate content under an ethical-cognitive design [21, 24].

7.5 Participatory Moderation (D5)

This dimension deals with the governance of conduct and community norms, regulating behavior rather than qualifying content (D3). It mobilizes *Groups*, in the facet of membership rules and protocols, and *Reputation*, as a behavioral indicator and restriction mechanism, whose management by *Groups* is directly affected by the degree of centralization of control [10].

The model postulates moderation in two complementary modes. Proactive moderation acts before an unproductive debate takes hold, configuring the discursive space through the explicit rules of the *Groups*, which make the presuppositions of the discussion visible in advance — an exercise of meta-communication in Angenot’s sense. Reactive moderation acts during the deterioration of the debate, mobilizing behavioral *Reputation* to sanction, flag, or contain escalation. The AI layer plays the role of sociotechnical mediator: instead of deciding autonomously, it summarizes histories, detects violations of *Groups*’ norms, and suggests reformulations, labelings, and referrals, reserving the final ethical decision for the human mediator [6, 28].

The operationalization of this governance presupposes not a single control center, but a distributed governance, in which normative responsibility is returned to the communities themselves, as observed on Bluesky and Reddit. This decentralization preserves community autonomy but carries a risk already evidenced in the comparative analysis: by defining their own boundaries, communities may close themselves into bubbles in which disagreeing groups mutually silence one another. This is not, however, an intrinsic flaw of the dimension, but the result of taking it in isolation: in the model, conceived to operate in an integrated manner, D5’s risk of fragmentation is counterbalanced by D4, which reintroduces external perspectives in a calibrated way, and by D2, which makes divergences visible rather than avoided. It is the composition of the five dimensions, and not the standalone application of any one of them, that sustains mitigation without sacrificing exposure to plurality.

7.6 The Transversal Generative AI Mediation Layer

Generative AI does not constitute a dimension of the model, but rather a transversal layer of sociotechnical mediation that cuts across all dimensions under the principle of human–AI symbiosis: AI takes on tasks of sensing, structuring, curation, exposure mediation, and governance support, reserving deliberation, critical thinking, and the final ethical decision for the human actor [8].

These roles specialize in each dimension — sensing in D1 and D5, structuring and synthesis in D2, curation and verification in D3, exposure mediation in D4, and governance decision support in D5 — but obey the same logic: expanding the cognitive capacities of participants without replacing human judgment.

8 Illustrative Application of the Model to an Empirical Case

In this section, we illustrate the application of the model to a real case: an interaction on the Instagram platform, selected because it concentrates, in a single high-visibility public post, Angenot’s three analytical categories. Usernames were anonymized in Figure 2, and offensive terms are reproduced verbatim solely because they are the very object of the discursive analysis.

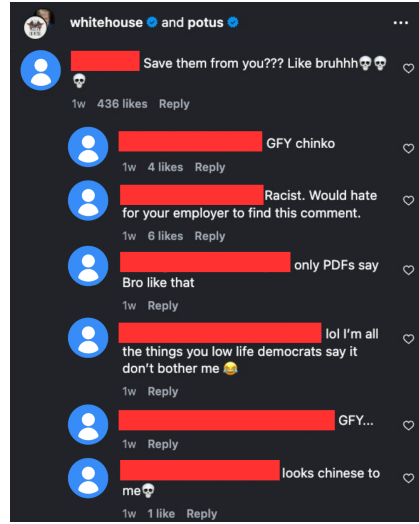


Fig. 2. Comment section of a post from the official White House profile in partnership with Donald Trump’s profile on the Instagram platform

8.1 Characterizing the Unproductive Debate

In the empirical case, communication fails not for lack of individual logic, but due to a structural condition: the participants operate in incompatible discursive universes. The following patterns were observed in the post in Figure 2:

- **Rhetoric of Incomprehension:** the debate immediately degenerates into the delegitimization of the interlocutor: the racial slur “chinko” reduces the opponent to a presumed ethnic category, shifting the confrontation from ideas to an attack on identity, and the slang term “PDFs” formulates, in coded language, a criminal accusation of pedophilia without any evidence.
- **Discursive Incommensurability:** the interlocutors operate with incompatible validity criteria: one side invokes a moral sanction (“it would be

a shame if your employer found this comment”), while the other neutralizes it by embracing the label as a symbol of political resistance (“failed democrats”), eliminating the common ground for validating evidence or values.

- **Illusion of Rationality:** the interactions constitute parallel monologues addressed to each side’s own audience: the statement that the insults “do not bother” them shows that the goal is to signal identity loyalty, not to exchange knowledge, breaking the externalization spiral.

8.2 Mapping the Model and Suggested Interventions

Applying the model’s dimensions makes it possible to project how design mechanisms and artificial intelligence could transform the nature of this interaction.

D1 – Interactional Deceleration: architectural frictions, such as minimum reading timers before posting, would break the immediate and aggressive reactivity observed in the comments, restoring the cognitive time needed for reflection and mitigating affective escalation.

D2 – Argumentative Structuring: replacing Instagram’s linear timeline with discussion trees would separate expressions of political support from personal accusations and rumors, making each side’s presuppositions visible.

D3 – Collaborative Curation: requiring sources for factual claims and an evidence repository, with AI verifying provenance and indexing historical contexts, would qualify the credibility of content before unsubstantiated accusations, such as those observed, could spread.

D4 – Mediated Exposure: instead of the abrupt dissemination of divergent perspectives, which generates resistance, AI would gradually introduce moderate counter-arguments, exposing participants to the internal coherence of the other’s system of thought, without aiming at direct persuasion.

D5 – Participatory Moderation: given the apparently absent moderation in the case, AI would act as a sociotechnical mediator, detecting offenses, hate speech, and patterns of unproductive debate, offering language reformulations before posting and flags to both users and human moderators.

The observed dysfunction thus stems from the engagement logic that characterizes the platform; the integrated application of the five dimensions, supported by Generative AI mediation, would make it possible to interrupt the *rhetoric of incomprehension* and reestablish the conditions for collective knowledge creation, demonstrating the model as a design instrument applicable to real digital scenarios.

9 Discussion

Returning to the research questions, the comparative analysis answered RQ1 by showing that alternative platforms offer robust mechanisms for debate structuring and systemic moderation, but rarely implement qualitative deceleration and divergence display — precisely the mechanisms most directly linked to the

meta-communication proposed on the basis of Angenot. In response to RQ2, the model articulates these findings into five mitigation dimensions cut across by a Generative AI mediation layer, offering a design vocabulary that connects the analytical categories of unproductive debate to concrete platform affordances.

The results reveal that the crisis of digital deliberation is not an isolated behavioral failure, but a direct consequence of the engagement logic characterized in Section 1, materialized in interfaces that stifle reflection and reward reactivity. From the perspective of collective knowledge creation, what occurs is a collapse of the knowledge spiral: without mutual trust and without a common ground of evidence, the processes of socialization and externalization become unfeasible.

A critical point is the role of Generative AI: while mainstream platforms use it as a “janitor” or binary exclusion filter, the model advocates its transition to governance infrastructure and sociotechnical mediation, under human–AI symbiosis — the technology replaces neither ethical judgment nor ideological dissent, but reduces cognitive overload by structuring informational chaos, allowing the debate to advance from the *rhetoric of incomprehension* toward qualified consensus or dissent.

There remains, however, an adoption tension: Kialo’s success in logically structuring debate contrasts with its low mass appeal, while Mastodon and Bluesky, despite returning sovereignty to users, inherit interface aesthetics that favor impulsiveness or bubbles. Mitigating polarization therefore requires a compromise between freedom of expression and deliberate frictions, suggesting that the design of sociotechnical systems must be, above all, pedagogical and ethical.

9.1 Implications for Platform Design and Governance

For platform designers, the model works as an instrument of analysis and generation: the five dimensions can be used both to audit existing systems, identifying, as in the comparative matrix, which mechanisms are absent, and to guide design decisions, such as introducing qualitative frictions (D1) or divergence visualizations (D2), which are rare today outside niches such as Kialo. For governance managers and moderators, the model suggests shifting from binary removal-based moderation to a two-mode mediation, proactive and reactive, in which AI flags and proposes reformulations, but the decision remains human. For researchers, the dimensions and classification criteria offer a replicable analytical vocabulary for comparative studies of deliberative affordances on other platforms.

9.2 Limitations and Future Work

This study has limitations that delimit the scope of its conclusions. First, the analysis of the platforms was based on public documentation, without direct observation of user behavior, which restricts inferences about the actual effectiveness of the mechanisms. Second, the application of the model is illustrative and based on a single case, serving as a demonstration of analytical plausibility rather than empirical validation. Third, the corpus of five platforms, although

theoretically diverse, does not exhaust the spectrum of existing architectures. Finally, the model presupposes an unresolved tension between deliberate frictions and the feasibility of large-scale adoption, evidenced by the contrast between Kialo’s structural rigor and its low mass appeal.

Future work should follow three directions: (i) the implementation of a prototype that instantiates the model’s dimensions on one of the analyzed platforms, prioritizing the Generative AI mediation layer; (ii) user studies to assess the impact of the interventions on the perceived quality of deliberation and willingness to dialogue, including the risks of cognitive overload and of perceived algorithmic paternalism; and (iii) the definition of operational metrics of debate productivity that enable comparative evaluation across platforms.

10 Conclusion

This work sought to analyze the problem of unproductive debates from the perspective of sociotechnical systems design, supported by KM principles and Generative AI. We observed that overcoming unproductive debates in the digital environment requires more than simply increasing the diversity of opinions. A profound reconfiguration of the sociotechnical infrastructures that sustain public dialogue is needed. The proposed model offers theoretical and practical foundations for designers and governance managers to operationalize digital spaces where ideological conflict does not degenerate into affective polarization and intellectual stagnation.

The main analytical contribution lies in the integration of the five mitigation dimensions with the transversal support of Generative AI: by transforming parallel monologues into explicit argumentative structures and calibrating exposure to divergent perspectives, the model seeks to restore the foundations of a productive digital democracy, with AI acting not as a tool of control, but as an ally in curation and in the preservation of collective memory. The structural re-design of social media is, in short, an imperative for the health of contemporary democracies.

Acknowledgments. During the preparation of this work, the authors used large language model-based AI to assist with text revision, restructuring of sections, and drafting of selected passages, which were subsequently reviewed and edited by the authors, who take full responsibility for the content of this publication.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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